Course Name: Operating System

Course Outcome

CO1-Understand the classification of operating system environment
CO2-Understand the basic of process management
CO3-Apply the concept of CPU process scheduling for the given scenarios
CO4-IIIlustration the process synchronization in OS
CO5-Analyze the occurrence of deadlock in OS
CO6-Describe and analyze the memory management
CO7-Understand the concept of disk scheduling

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Carryover Examination, Even Semester, 2023-24
B.Tech (CS), II, III
BCSC 0004 & OPERATING SYSTEM

Time: 3 Hours **Maximum Marks: 80**

Section – A

*Attempt **All** Questions* 3 X 8 = 24 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Discuss the role of the kernel in an operating system and its interaction with user-space processes.	3	1	U	F
2	Explain the purpose of system calls in operating systems. Give examples of common system calls.	3	2	R	C
3	Describe the potential issues that may arise when multiple processes concurrently access a shared resource without synchronization.	3	1	R	P
4	Provide a step-by-step explanation of how Peterson Solution based synchronization can be implemented to coordinate the execution of processes P1 and P2, ensuring mutually exclusive access to the printer.	3	3	A	C
5	How does the Banker's algorithm prevent deadlock in resource allocation?	3	4	U	D
6	Provide examples of real-world scenarios where deadlock might occur in an operating system environment.	3	2	E	S
7	Consider a disk with 200 tracks (numbered from 0 to 199) and the disk arm is currently positioned at track 50. Given the following sequence of track requests: 82, 170, 43, 140, 24, 16, 190. Calculate the total head movement using the SSTF (Shortest Seek Time First) disk scheduling algorithm.	3	4	R	F
8	Assume a system with 16KB of physical memory and a page size of 2KB. If the virtual address space is 32KB,	3	5	U	S

	calculate the number of bits required for the page number, offset, and frame number in a two-level paging system.				

Section – B

*Attempt **All** Questions* 6 X 6 = 36 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain the concept of process management in an operating system. Describe the life cycle of a process and discuss the role of process control block (PCB).	6	2	U	C
2	Provide examples of scenarios where multi-threading within a process might be advantageous in an operating system environment.	6	3	A	F
3	Describe the concept of process synchronization and its importance in multi-process environments. **OR** What are the components of a process control block (PCB)?	6	2 2	U U	P C
4	A computer system has 4 frames in its physical memory and uses the FIFO (First-In-First-Out) page replacement algorithm. Given the following reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5. Determine the number of page faults generated by the FIFO algorithm.	6	2	An	D
5	A single processor system has three resource types X, Y and Z, which are shared by three processes. There are 5 units of each resource type. Consider the following scenario, where the column alloc denotes the number of units of each resource type allocated to each process, and the column request denotes the number of units of each resource type requested by a process in order to complete execution. Which of these processes will finish LAST?	6	2	E	C
6	Discuss the role of TLB (Translation Lookaside Buffer) in virtual memory management. OR	6	3	A	S

	Alloc			**Request**		
	X	**Y**	**Z**	**X**	**Y**	**Z**
P0	1	2	1	1	0	3
P1	2	0	1	0	1	2
P2	2	2	1	1	2	0

	Describe the mechanisms used by modern operating systems to support large address spaces and 64-bit memory addressing.		4	U	P

Section – C

*Attempt **All** Questions* 10 X 2 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	In a multiprogramming environment, multiple processes are running concurrently. Suddenly, one process monopolizes the CPU, causing other processes to starve. a. Identify the possible reasons for CPU starvation in this scenario. b. Discuss the role of process scheduling algorithms in preventing CPU starvation. c. Propose strategies to mitigate CPU starvation and ensure fair process execution. **OR** Consider the set of 5 processes whose arrival time and burst time are given below- **Process Id / Arrival time / Burst time** **P1 / 3 / 1** **P2 / 1 / 4** **P3 / 4 / 2** **P4 / 0 / 6** **P5 / 2 / 3** If the CPU scheduling policy is SJF preemptive, calculate the average waiting time and average turnaround time.	10	2 4	U A	C PC
2	How does a file system organize data on storage devices, and what are its primary objectives in doing so? **OR** Consider a disk queue with requests for I/O to block on cylinders 98, 183, 37, 122, 14, 124, 65, 67. If the disk head is started 53 find out the total head movement with respect to FCFS, SSTF, SCAN, C-SCAN and LOOK scheduling.	10	4 3	C A	D C